

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

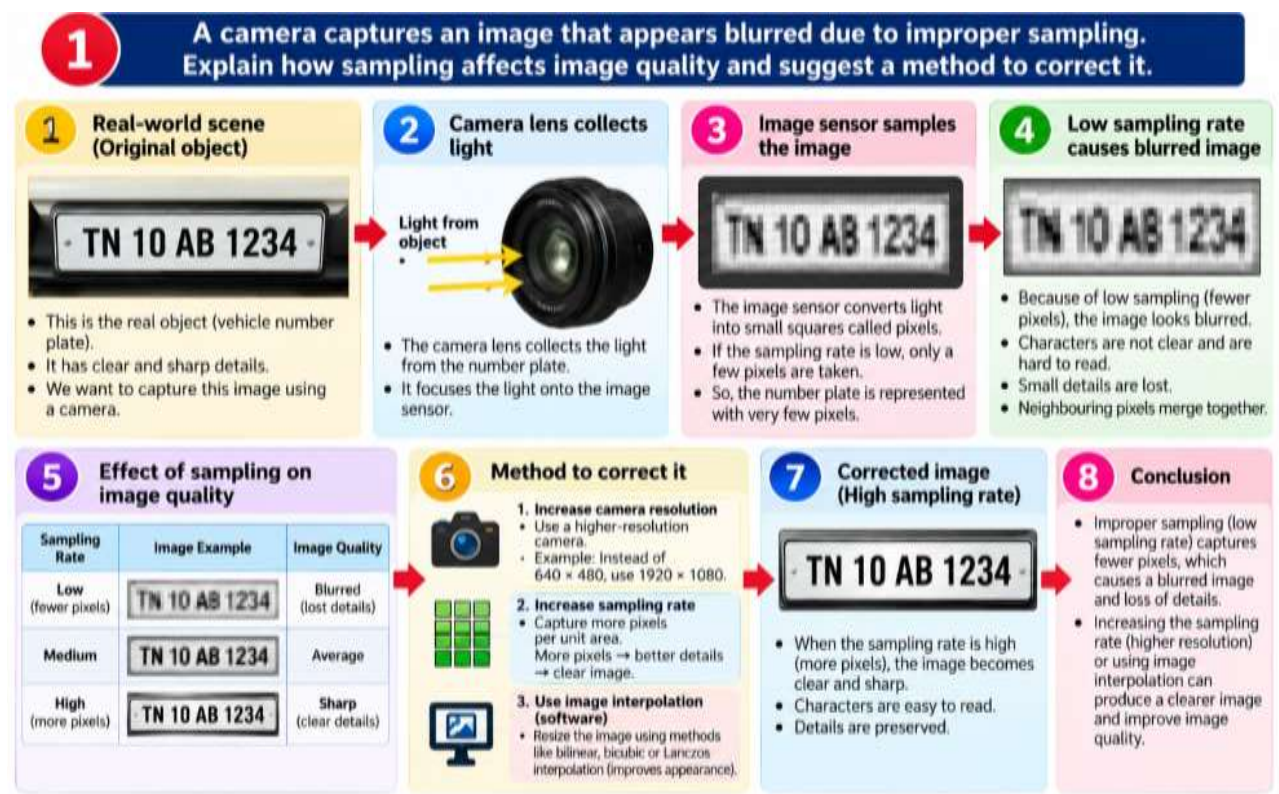
COURSE NAME: COMPUTER VISION

COURSE CODE:ITA0521

COURSE OUTCOME COVERED

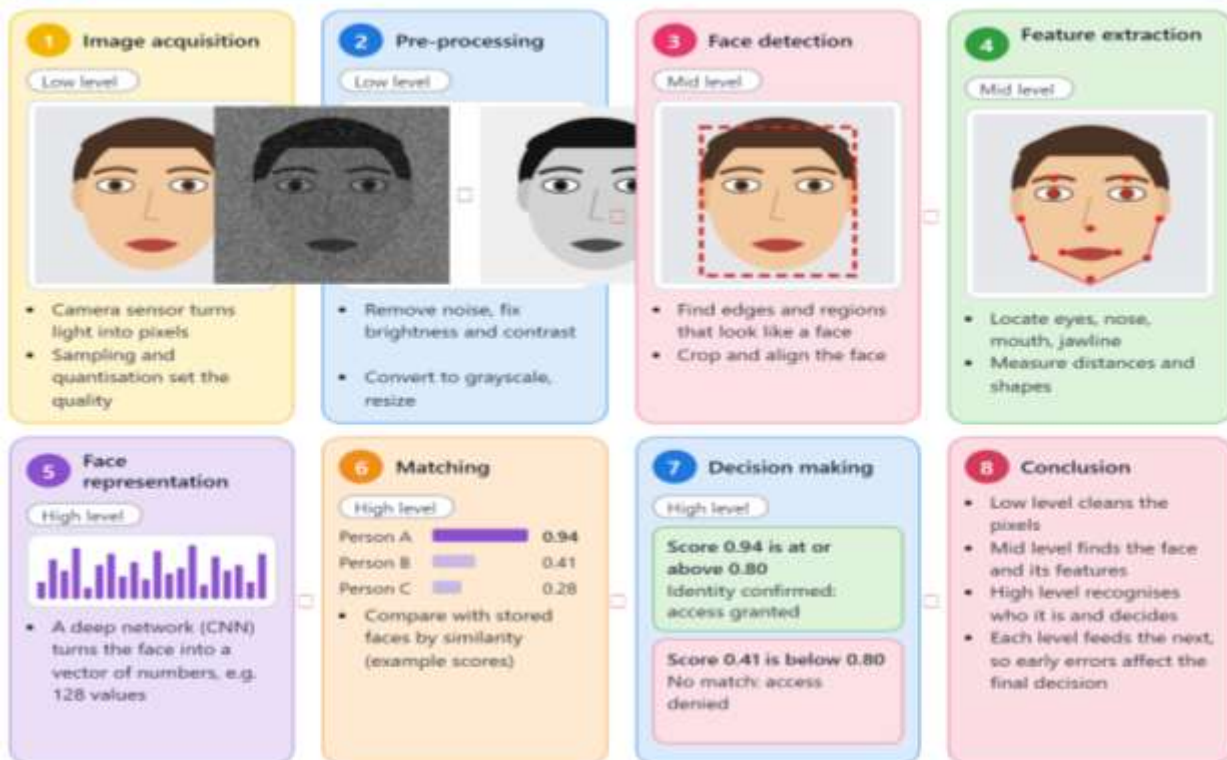
CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)

Assessment Tool Weightage: 40%



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In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.



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A surveillance image appears noisy in low light. Analyze how sensing and acquisition conditions contribute to noise.



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An image appears overly smooth with loss of detail. Explain how quantization levels contribute to this issue.

1 Original scene (256 levels)



- Brightness varies continuously
- Fine texture and gentle shading are present

2 Quantization step

Continuous brightness

Round to the nearest of $L = 2^b$ levels

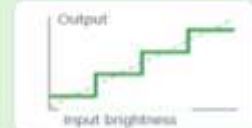
Stored as a b-bit code

- Step size = range / L
- Rounding error is up to half a step

3 Bit depth sets the levels



4 Staircase mapping (4 levels)



- Many inputs share one output
- Dashed line is the true value

5 Visible effect



16 levels: faint banding



4 levels: flat, posterized

6 Why detail disappears

Original pixels

118 121 124 127

After 4-level quantization

85 85 85 85

- Small differences merge into one value
- Texture flattens; slow gradients break into bands

7 How to correct it

- Capture at higher bit depth (8, 10, 12 bit)
- Dithering to hide banding
- Non-uniform quantization with finer steps where the eye is sensitive
- Histogram equalization to use the levels better

8 Conclusion

- Too few quantization levels merge nearby intensities
- The image looks smooth, flat and banded
- Lost detail cannot be recovered afterwards
- More levels (higher bit depth) preserve detail

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In a facial recognition system, explain how resolution impacts feature extraction accuracy.

1 Captured face image



- Resolution = pixels covering the face
- It sets how much detail the camera records

2 What feature extraction needs

Landmarks: eye corners, nose tip, lip edges

Shape: jawline, brow ridge, cheeks

Texture: skin, moles, wrinkles

Embedding: a CNN number vector built from all of these

3 High resolution



- Sharp edges, exact landmarks
- Rich texture gives a distinctive embedding

4 Low resolution

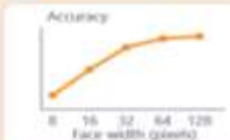


- True point (green dot), Detected point (red dot)
- Landmarks drift, texture is lost

5 Same face, four resolutions



6 Accuracy vs resolution



- Steep gain at low resolution, then it levels off
- Illustrative trend, not measured data

7 How to improve it

- Higher-resolution camera or closer, well-framed capture
- Good focus and lighting
- Face super-resolution before extraction
- Train on low-resolution samples
- Fuse several video frames

8 Conclusion

- Resolution decides how much facial detail reaches feature extraction
- Too low: landmarks drift and embeddings become less distinctive, so errors rise
- Beyond a point, extra pixels add little

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A vision system fails in low-light conditions. Explain how image sensing impacts system performance.

1 Vision system at night



- Detection and recognition rely on the sensor image
- At night the sensed image is poor

2 The sensing chain

Lens, aperture: collect light

Photodiode: photons to electrons

Amplifier: applies gain

ADC: analogue to digital

3 Weak signal, low SNR



SNR, illustrative

- Fewer photons, and SNR grows only as their square root
- Small pixels, slow lens and short exposure lose more light

4 Noise and limited range

Shot noise: random photon counts

Dark current: heat-made electrons

Read noise: electronics errors

Gain: brightens, but amplifies noise too

Dynamic range: shadows crushed, lamps blown out

5 Sensed image



Daylight: clear



Low light: dark and grainy

6 Edge detection result



Day: clean outlines



Night: speckle, weak edges

7 How to improve it

- Larger sensor, bigger or back-illuminated pixels
- Wider aperture, longer exposure
- IR illumination or a thermal camera
- Denoising and low-light enhancement
- Multi-frame or HDR capture
- Train on night images, fuse radar or LiDAR

8 Conclusion

- Sensing sets the quality of everything downstream
- Low light gives low SNR and extra noise
- Edges, features and detections then fail
- Fix the sensing first, then the software

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In medical imaging, explain how image acquisition affects diagnostic reliability.

1 Acquisition modalities

X-ray, CT: X-ray attenuation

MRI: radio signals from hydrogen

Ultrasound: returning echoes

PET: tracer gamma rays

- Each one senses the body differently

2 Acquisition settings

Spatial resolution: pixel and slice size

Contrast: bit depth, windowing

SNR: dose, exposure, averaging

Positioning: field of view, patient alignment

Scan time: how long the patient must stay still

3 Trade-offs

Dose $\square$ noise
Less radiation, grainier image

Resolution $\square$ SNR
Smaller voxels, weaker signal

Scan time $\square$ motion
Longer scans, more movement

4 Artifacts introduced

Motion: breathing, heartbeat

Noise: low dose or short scan

Metal streaks: implants

Partial volume: thick slices blur small lesions

Calibration drift: wrong intensities

5 Can you still see the lesion?



Good acquisition



Low dose



Low resolution



Patient motion

Red ring marks a small nodule

6 Effect on diagnosis

- Missed lesions (false negatives)
- Artifacts mistaken for disease (false positives)
- Wrong size measurements
- AI tools misread the image

Sensitivity, illustrative



7 Improving reliability

- Standard, protocol-based scanning
- Regular calibration and phantom QA
- Careful positioning and immobilisation
- Breath-hold or gating for motion
- Thin slices for small lesions
- Dose set by ALARA, plus reconstruction and denoising
- Metal artifact reduction

8 Conclusion

- Acquisition sets resolution, contrast and noise
- Poor acquisition hides real findings or creates false ones
- Software cannot recover what was never captured
- Good acquisition is the base of a reliable diagnosis

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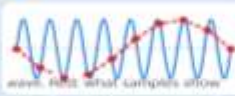
A digital image appears overly compressed. Explain how sampling and quantization contribute to information loss.

1 Continuous scene



- Real scenes vary smoothly in space and brightness
- A computer must store a finite number of values

2 Sampling (where)



- Picks values on a pixel grid
- Too few samples lose fine detail and cause aliasing

3 Quantization (how much)



- Rounds each brightness to one of $L = 2^b$ levels
- Few levels merge nearby tones and cause banding

4 Size = $M \times N \times b$ bits

512 x 512, 8-bit
256 KB

128 x 128, 8-bit
16 KB (fewer samples)

128 x 128, 3-bit
6 KB (fewer levels too)

- Smaller file, but information is thrown away

5 Seeing the loss



ABC 123 xyz
Original



ABC 123 xyz
Sampled only (32 x 32)



ABC 123 xyz
Quantized only (4 levels)



ABC 123 xyz
Both: overly compressed look

6 What is lost

Sampling: fine edges, small text, texture; false patterns (aliasing)

Quantization: gentle gradients, subtle contrast; bands and false contours

Together: blocky, flat, posterized

Irreversible: the original values cannot be recovered

7 How to avoid it

- Sample at least twice the finest detail (Nyquist)
- Low-pass filter before downsampling
- Use more bits per pixel (8 or more)
- Dithering to hide banding
- Keep lossless masters (PNG, TIFF)
- Interpolation can smooth the look but cannot restore detail

8 Conclusion

- Sampling limits spatial detail
- Quantization limits tonal detail
- Too little of either discards information for good
- Choose resolution and bit depth for the detail you need

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A vision system detects objects incorrectly due to poor acquisition. Explain the relationship between sensing and detection accuracy.

1 Scene to detect



- A car, a sign and a pedestrian
- The detector must find and label each one

2 Sensing quality factors

Resolution: pixels on the object

Exposure, light: brightness and contrast

Focus, motion: sharpness of edges

Noise (SNR): clean or grainy

Dynamic range: shadows and highlights

3 Errors flow downstream

Sensing (camera image)

Features (edges, texture, shape)

Detector (class and box)

Output (found or missed)

4 Damage to features

Blur: edges weaken, shapes merge

Noise: false edges appear

Low contrast: object blends into background

Low resolution: small objects get too few pixels

Clipping: detail lost in dark or bright areas

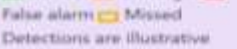
5 Same scene, four acquisitions



Good



Low light



Blur



Low resolution

- Correct (green box)
 - Off target (orange box)
 - False alarm (red box)
 - Missed (yellow box)
- Detections are illustrative

6 Quality vs accuracy



- Accuracy rises with sensing quality, then levels off
- Failures: missed, false, mislocated or mislabelled objects

7 How to improve it

- Better sensor and lens, proper lighting
- Auto exposure, fast shutter, stabilisation
- Enough resolution for the smallest target
- Denoise and deblur before detection
- Train with degraded images
- Fuse other sensors (IR, radar, LiDAR)

8 Conclusion

- Detection can only use what sensing captures
- Poor sensing weakens the features the detector relies on
- Better algorithms help only up to that limit
- Improve sensing first

1 Part under inspection



- A metal surface with four defects: scratch, pit, crack, stain

2 Pixel size on the part

Pixel size = field of view / pixels

$$100 \text{ mm} / 1000 \text{ px} = 0.1 \text{ mm per pixel}$$

A defect should span 2 to 3 pixels or more (Nyquist)

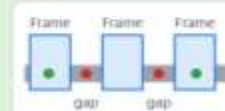
So the smallest reliable defect is about 0.25 mm

3 Spatial sampling



- Fine grid; the line fills whole pixels and stands out
- Coarse grid; it is averaged with background and fades

4 Temporal sampling



- Frame rate too low for belt speed leaves gaps
- Defects in a gap are never imaged

5 Detection at four pixel sizes



Fine: 4 of 4 found



Medium: 4 of 4 found



Coarse: 3 of 4 found



Very coarse: 2 of 4 found

Found Missed red = flagged pixels

6 Smallest detectable defect

0.05 mm px	0.13 mm
0.1 mm px	0.25 mm
0.2 mm px	0.5 mm
0.4 mm px	1.0 mm

About 2.5 pixels wide. Coarser sampling means lower sensitivity.

- Averaging dilutes a small defect's contrast below the threshold

7 How to improve it

- Higher-resolution or line-scan camera
- Smaller field of view, or several cameras
- Frame rate matched to belt speed
- Strobe light and short exposure
- Lighting that raises defect contrast
- Sub-pixel methods; zoom on critical areas
- Trade-off: more data, slower, less light per pixel

8 Conclusion

- Sampling sets the smallest defect the system can see
- Coarse spatial sampling dilutes small defects
- Slow temporal sampling skips defects entirely
- Match sampling to the smallest defect and the belt speed